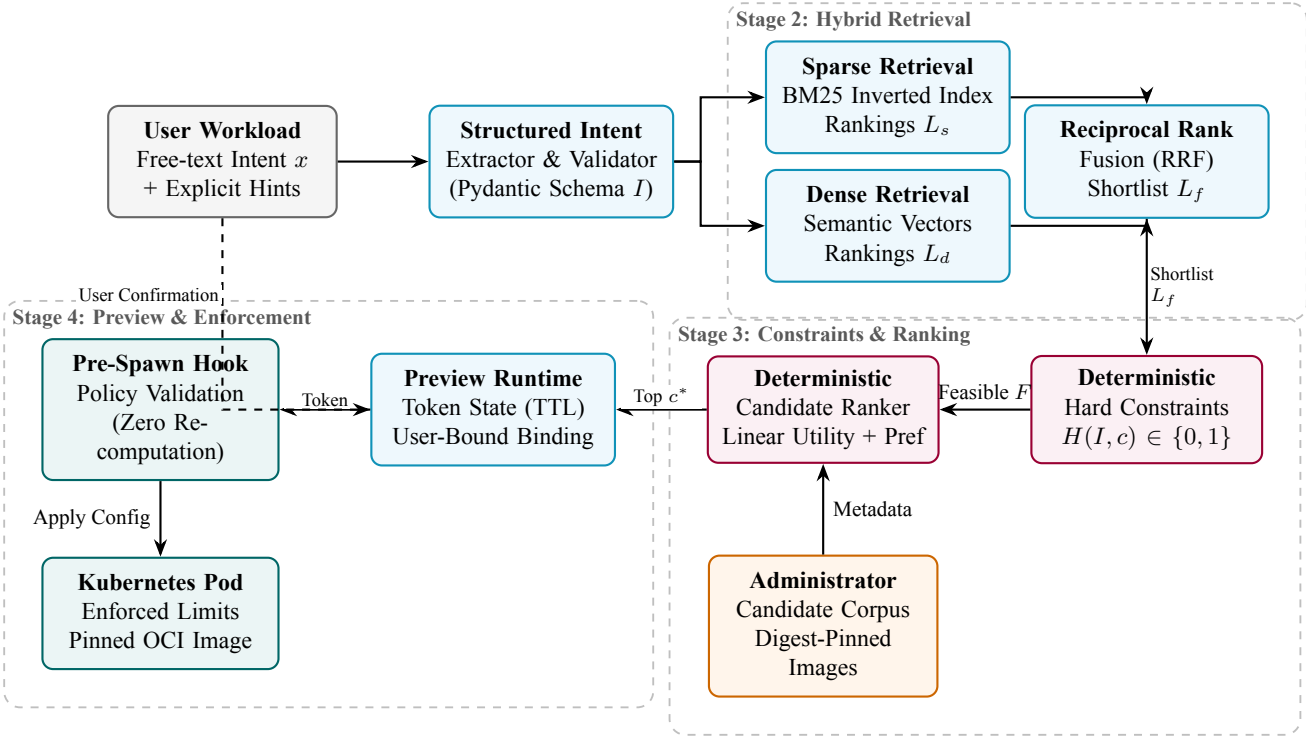

Stage 2: Hybrid Retrieval
User Workload
Free-text Intent x
+ Explicit Hints
Structured Intent
Extractor & Validator
(Pydantic Schema I)
Sparse Retrieval
BM25 Inverted Index
Rankings L_s
Dense Retrieval
Semantic Vectors
Rankings L_d
Reciprocal Rank
Fusion (RRF)
Shortlist L_f
Shortlist
L_f
User Confirmation
Stage 4: Preview & Enforcement
Pre-Spawn Hook
Policy Validation
(Zero Re-
computation)
Token
Preview Runtime
Token State (TTL)
User-Bound Binding
Top c^*
Stage 3: Constraints & Ranking
Deterministic
Candidate Ranker
Linear Utility + Pref
Feasible F
Deterministic
Hard Constraints
$H(I,c) \in \{0,1\}$
Apply Config
Kubernetes Pod
Enforced Limits
Pinned OCI Image
Metadata
Administrator
Candidate Corpus
Digest-Pinned
Images